12.10 Tying Loose Ends

WORKSHEET

The question

STEP III 2008, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q13

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself After tying two ends together, how many pieces of string are left (counting a ring as no longer a "piece")? Does it depend on whether a ring was made?

STAGE 2 All the knots

Ask yourself By the observation above, what is the situation before the second knot? The third? Let I_k indicate a ring at the knot made when k pieces remain. What are the $\mathrm{E}(I_k)$?

STAGE 3 The variance

Ask yourself Are the indicators I_k independent? (Does the probability of a ring at one step depend on what happened earlier?)

STAGE 4 Forty thousand strings

Ask yourself Write the sum of the reciprocals of the odd numbers up to $2n-1$ using harmonic numbers.

Try this next

1. **Small cases.** For $n = 2$, list the possibilities and check that the expected number of rings is $\frac{4}{3}$.

2. **One big loop.** Show that the probability that the process ends with a single loop containing all the strings is $\prod_{k=2}^{n}(1 - \frac{1}{2k-1})$, and that this is approximately $\sqrt{\frac{\pi}{4n}}$ for large n. (*Hint:* relate the product to $\binom{2n}{n}$.)

3. **Cycles of permutations.** Show that the expected number of cycles of a random permutation of n objects is H_n, by building the permutation one element at a time (the "Chinese restaurant" construction).

4. **The variance.** Show that $\mathrm{Var}(R) = \mathrm{E}(R) - \sum_{k=1}^{n} \frac{1}{(2k-1)^2}$, and that the subtracted sum tends to $\frac{\pi^2}{8}$.

Further reading

- **Peter Winkler, *Mathematical Puzzles: A Connoisseur's Collection*** (A K Peters), which includes the string-tying puzzle.
- **Richard Arratia, Andrew Barbour & Simon Tavaré, *Logarithmic Combinatorial Structures*** (European Mathematical Society). Cycles, primes and random structures, for later.
- **Colin Adams, *The Knot Book*** (American Mathematical Society). An introduction to knot theory, including random knots.

The full guided solution, and what the question leads to at university, is in Section 12.10:
https://beyondstep.keithmcnulty.org/ch12/loose-ends.html